Adam Boesky

Astrophysics PhD Student and NSF Graduate Research Fellow, Harvard University
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As of 2026-11-14[1], published 4 refereed papers (3 first-author) with 94 total citations, and an h-index of 4; a complete listing can be found on ADS.

NSF Graduate Research Fellow (2026); Finalist for the Hertz Fellowship, Rhodes Scholarship, and Marshall Scholarship (2025).

Lead developer of *SPLASH* and *ztforce*; contributor to *COMPAS* and *Prost*; a full list can be found on GitHub.

Experience

Center for Astrophysics | Harvard & Smithsonian, Cambridge, MA

Aug 2023 – present *Researcher, Villar Time-Domain Astronomy Data Lab*

Jun 2022 – present *Researcher, Berger Cosmic Transient Group*

Leading the first multi-survey search for decade-long transients: compiled a catalog of candidates and am now characterizing their physical nature. Developed, released, and maintain *SPLASH*, a machine learning pipeline that classifies supernovae from the inferred properties of their host galaxies at ∼500 objects per second, built for the alert stream of the Rubin Observatory's LSST (first-author, *ApJS* 2026). Ran binary population synthesis simulations with *COMPAS* to model the cosmic merger rates of black holes and neutron stars, showing that they deviate by up to $10\times$ from commonly assumed star-formation-based models (two first-author *ApJ* papers, 2024). Contributed to *Prost*, a probabilistic host-galaxy association code, and *RESOLVE*, a surrogate likelihood for gravitational-wave parameter estimation.

SpaceX, Los Angeles, CA

Summers 2023, 2024 *Guidance, Navigation, & Control Engineering Intern*

Developed physical and empirical models for high-fidelity Monte Carlo simulation of the Falcon 9 and Falcon Heavy rockets. Drove operational changes to improve vehicle reliability and performance. Automated day-of-launch operations, built tools to accelerate launch cadence, and assessed the value of mission-specific launch analyses.

ProDex Labs, New York, NY

Mar 2025 – Mar 2026 *Founding Engineer*

Developed a discrete-event simulation engine for building digital twins of manufacturing processes. Designed and implemented the planning and scheduling optimizers. Integrated an AI agent into the application, speeding up user analysis by $> 100\times$. Led development of several modules and managed teams of engineers.

Harvard School of Engineering & Applied Sciences, Cambridge, MA

Jun 2020 – Aug 2022 *Research Assistant, C3 Policing Research Team*

Quantified the impact of the Counter Criminal Continuum (C3) policing method across business, crime, education, health, and housing data; presented results to Massachusetts officials.

Education

PhD in progress (2026 – present), Astrophysics, Harvard University.

BA 2025, Astrophysics and Mathematics, Harvard University, *cum laude*.

Talks, Teaching & Service

Talks at Orange County Astronomers (2026), APS April Meeting (2024), and Post-Pax-VIII (2022); poster at AAS 245 (2025).

Teaching assistant for Harvard Physics 143a and 12b; research mentor to three high school students, including a 2026 Regeneron Science Talent Search Finalist; referee for *The Astrophysical Journal*.

[1] Publication data collected from NASA Astrophysics Data System